

Beyond Recall: Operational Correctness for Agent Memory Systems

OpMem v0 Benchmark and Empirical Pins

Brainy Research

September 18, 2026

Abstract

Retrieval benchmarks such as LoCoMo and LongMemEval measure whether an agent recalls the right fact, but not whether a memory *system* honors forget, correct, isolate, and supersede semantics in production. We present **OpMem v0**, a 13-task, pass/fail suite over neutral **remember/recall/revise/forget** scripts with substring-based assertions on ranked results. On the frozen 2026-08-15 pin (git SHA `1b5ab3e`), Brainy passes **13/13** via `GET /memories/search` while Mem0 **Platform** passes **10/13** (failures: `cor02`, `sup03`, `upd02`). This manuscript reports repository evidence, a reproducible protocol, threats to validity, and new embedded-harness re-measurements on the current merge-pin git revision where completed. We do *not* claim conversational-recall superiority over Mem0; OpMem operational leadership is orthogonal to LoCoMo J-scores.

1 Introduction

Production agents fail when suppressed memories leak after re-ingestion, corrections do not stick, tenants bleed across isolation boundaries, or stale facts outrank updates. Conversational QA benchmarks [3, 5] do not encode these lifecycle contracts; hallucination-oriented memory benchmarks such as HaluMem [2] focus on factual faithfulness rather than governed delete/correct semantics.

OpMem targets **operational correctness**: deterministic scripts, binary scoring, and explicit adapter mappings to each system’s native API surface [1]. Section 8 lists threats; Section 6 separates **completed** evidence from **blocked** external comparisons.

2 Related work

Retrieval QA. LoCoMo [3] and LongMemEval [5] score end-to-end answer quality over long conversations; they are not substitutes for lifecycle tests.

Memory products. Mem0 documents accumulation-oriented designs and names staleness as an open problem [4]. We report Mem0 **Platform** numbers only on OpMem fixtures; we do not import Mem0 blog LoCoMo percentages as same-pin scores.

Pending external comparisons. Paid or vendor-hosted counter-runs are allowed only after a written cost estimate and explicit approval; that gate was not a categorical denial of spend. Mem0 Platform on OpMem v0 is **completed** at negligible API volume (13 tasks, 47 scripted steps; see Section 6.8). Mem0 OSS, Zep, Letta, and LangMem remain **blocked** because no OpMem adapter, access credentials, or line-item cost worksheet exists yet (Section 6.9).

3 Neutral operation model

Actors are (tenant, subject) pairs. Tasks are operation scripts; `revise` and `forget` reference prior `remember` step ids. Adapters namespace tenants per task (`opmem-{nonce}-{task}`) for hermetic runs.

Table 1: Adapter mappings (normative; from `PUBLICATION_READINESS.md`).

Op	Brainy	Verbatim baseline	Mem0 Platform
remember	POST /ingest	append	POST /v1/memories/
recall	GET /memories/search	token overlap	POST /v2/memories/search/
revise	POST .../correct	in-place replace	PUT /v1/memories/{id}/
forget	POST .../suppress	delete	DELETE /v1/memories/{id}/

4 Task suite

OpMem v0 comprises 13 JSON fixtures under `fixtures/opmem/`: three suppression, three correction, three isolation, three staleness (including `upd03_state_supersession`), and one idempotency task. Assertions use case-insensitive substring checks on ranked `content` fields (`min_results`, `top_contains`, `must_exclude`, etc.).

5 Methodology

5.1 Scoring lane

Primary pins use Brainy `/memories/search`, not product `POST /recall` (Section 6.4 reports a `/recall` lane ablation). The harness (`evals/run_opmem.py`) exits non-zero only on infrastructure errors, not task failures—CI proves runner health, not 13/13, unless JSON is parsed.

5.2 Reproduction commands

```
go test ./internal/api/ -run TestOpMemBenchmarkAgainstHTTPServer -count=1
python3 evals/run_opmem.py --systems verbatim,brainy \
  --base-url http://127.0.0.1:8080 --json-out opmem-report.json
```

Paper experiments: `scripts/run-opmem-manuscript.sh` and `evals/run_opmem_paper_experiments.py`.

6 Results

6.1 Frozen 2026-08-15 pin (historical, SHA 1b5ab3e)

[completed] Brainy **13/13**; verbatim **10/13**; Mem0 Platform **10/13** (artifacts `opmem-fresh-local-20260815.m`
`opmem-mem0-fresh-20260815.md`).

Mem0 failures on that pin: `cor02` (correction stickiness), `sup03` (durable forget leak), `upd02` (preference ranking).

Table 2: Category breakdown on 2026-08-15 Mem0 Platform counter-run.

Category	Mem0 Platform	Brainy (same cycle)
correction	2/3	3/3
isolation	3/3	3/3
suppression	2/3	3/3
staleness	2/3	3/3
idempotency	1/1	1/1
overall	10/13	13/13

6.2 Current-SHA re-measurement

[**completed**] At git SHA 20533f2ca83e (UTC 2026-09-18T23:12:29Z), embedded API harness: Brainy (search lane) **13/13**, verbatim **10/13**. Mem0 Platform (same run): **10/13** (API surface: Platform via Mem0OpAdapter).

6.3 Combined machine-readable artifact

[**completed**] Committed `opmem-primary.json` lists per-task outcomes for verbatim, brainy, mem0 with manifest fields (SHA, date, fixture count).

6.4 Recall vs search lane

Table 3: Brainy recall lane and revise mapping (SHA 20533f2ca83e).

Configuration	Overall
brainy	13/13
brainy-recall	9/13
brainy-supersede	11/13

Search (`/memories/search`) is the normative pin lane. `brainy-recall` uses `POST /recall` (context mode). Correction tasks on the recall lane may return HTTP 409 during `/correct` (infrastructure errors in artifact JSON).

6.5 Correct vs supersede mapping

[**completed**] Mapping `revise` $\rightarrow$ `POST .../supersede` yields **11/13** vs **13/13** with `/correct` on the search lane (same SHA). Failures are task-level assertion misses, not missing runs.

6.6 Mechanism breakdown

Category pass rates separate correction/suppression/staleness contracts; verbatim approximates rank-only recall without governed lifecycle ops.

6.7 Statistical stability

[**completed**] Five consecutive Brainy search-lane repeats on one embedded server session. **Valid runs only** (infra errors = 0): 2/5; scores: 13/13.

Category	Brainy	Verbatim (rank-only)
correction	3/3	2/3
idempotency	1/1	0/1
isolation	3/3	3/3
staleness	3/3	3/3
suppression	3/3	2/3

- Repeat 1: 13/13 (infra errors=0)
- Repeat 2: 9/10 (infra errors=3)
- Repeat 3: 9/10 (infra errors=3)
- Repeat 4: 13/13 (infra errors=0)
- Repeat 5: 9/10 (infra errors=3)

Repeats with infra errors > 0 (3 of 5) are **invalid/incomplete**: the harness omits errored tasks from the denominator, so aggregates such as 9/10 are *not* comparable to a clean 13/13 pin. Use isolated `TestOpMemBenchmarkAgainstHTTPServer` for merge-gate stability.

6.8 External systems

[**completed**] **Mem0 Platform**: 10/13 on SHA 20533f2 (`opmem-pin-20260918.json`); same headline as 2026-08-15 pin. Completed under negligible Platform API use (full suite ≈ 47 scripted steps). [**blocked**] **Mem0 OSS**, **Zep**, **Letta**, **LangMem**: blocked pending adapter + access + cost worksheet (Section 6.9); not run on this pass.

6.9 Cost-estimation plan (before any new paid run)

OpMem v0 has 13 tasks and 47 scripted steps (`fixtures/opmem/`). For each pending system, prepare billing before requesting approval:

- **Mem0 OSS** — Implement `Mem0OSSOpAdapter` (version-pinned `mem0ai/mem0`); map neutral ops to OSS HTTP/SDK. *Estimate*: count per-step Platform-equivalent calls (remember/re-call/revise/forget); self-hosted OSS = engineer time + local compute only; hosted OSS = vendor rate card $\times$ call count (prepare spreadsheet from public pricing when deployment target is chosen).
- **Zep** — Implement Zep adapter (add/search/update/delete on Zep Cloud or self-hosted Graphiti). *Estimate*: same 47-step script $\times$ mapped Zep API units; obtain unit prices from Zep billing docs/dashboard before run.
- **Letta** — Implement Letta memory/agent API adapter. *Estimate*: 47 steps $\times$ Letta memory CRUD + search calls; quote from Letta Cloud pricing for memory operations.
- **LangMem** — Clarify target surface (LangGraph store vs hosted LangMem product); implement adapter. *Estimate*: primarily local/library compute if OSS; if hosted, count store read/write units per step.

No dollar totals are asserted here until rate cards and deployment mode are pinned; approval requires the completed worksheet, not a categorical spend ban.

7 Discussion

OpMem failures on Mem0 align with accumulation-oriented update semantics: corrections and durable forgets are not first-class lifecycle events. Verbatim baseline passes 10/13, showing the suite discriminates rank-only stores from governed suppression/correction. Policy disagreements remain (sup03 durable forget is a normative contract).

8 Limitations

1. Small $n = 13$; each task is $\sim 7.7\%$ of the aggregate score.
2. Substring scoring does not guarantee semantic safety.
3. Search vs `/recall` product path may diverge (lane ablation quantifies on Brainy).
4. English, domain-neutral fixtures; no concurrency or TTL tests.
5. Mem0 Platform API drift; OSS/Zep/Letta/LangMem not measured until adapter + cost approval.
6. Fixture authorship bias; no independent external audit documented.

9 Artifact appendix

Committed run manifests and JSON under `docs/benchmarks/artifacts/opmem-manuscript-*`: `run-manifest.json`, `opmem-primary.json`, `opmem-lane-ablation.json`, `opmem-stability.json`, `opmem-mechanism.json`, and `commands.log`.

Build PDF: `make -C docs/research/opmem/paper pdf`.

10 Future work

v1 task expansion (~ 30 tasks), Zep/Letta/LangMem adapters, supersede-native revise mapping as default where lineage is required, and fail-closed CI assertion of 13/13 on merge pins.

References

- [1] Brainy. OpMem v0 specification and harness. <https://github.com/tryvinci/brainy/docs/research/opmem-spec.md>, 2026.
- [2] Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchu Li, Feiyu Xiong, and Zhiyu Li. HaluMem: Evaluating hallucinations in memory systems of agents. *arXiv preprint arXiv:2511.03506*, 2025.
- [3] Adyasha Maharana et al. LoCoMo: Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17710*, 2024.
- [4] Mem0. State of AI agent memory (2026). <https://mem0.ai/blog/state-of-ai-agent-memory-2026>, 2026. Vendor context on staleness and cross-session evolution; not an OpMem same-pin score.

- [5] Howard Yen Wu et al. LongMemEval: Benchmarking long-term memory in llm chat assistants. *arXiv preprint arXiv:2410.10819*, 2024.